SECTION 8: **List of Courses and No of Units by Levels in tabular form**

100L FIRST SEMESTER			
Course Code	**Course Title**	**Status C or E**	**Units**
GEG 113	Engineering Applied Mathematics I	C	3
GEG 115	Engineering Algebra I	C	2
GEG 117	Engineering Calculus I	C	2
MEG 111	Workshop Practice I	C	1
FSC 112	Introductory Chemistry I	C	3
FSC 115	Introductory Physics I	C	3
GST 102	Philosophy, Logic and Philosophy of Science		2
GST 105	Use of English	C	2
Total Units of Compulsory Courses			18
Total Units of Elective Courses			0

100L SECOND SEMESTER			
Course Code	**Course Title**	**Status C or E**	**Units**
GEG 124	Engineering Applied Mathematics II	C	3
GEG 126	Engineering Algebra II	C	2
GEG 128	Engineering Calculus II	C	2
MEG 121	Workshop Practice II	C	1
PHS 121	Introductory Physics II	C	2
PHS 122	Introductory Physics III	C	3
PHS 123	Physics Practical	C	2
GST 103	Nigerian peoples and Cultures I	C	2
Total Units of Compulsory Courses			17
Total Units of Elective Courses			0

200L FIRST SEMESTER			
Course Code	**Course Title**	**Status C or E**	**Units**
GEG 217	Engineering Calculus III	C	2
GEG 219	Ordinary Differential Equations	C	2
SSG 215	Introduction to Engineering Computing	C	2
EEG 213	Signals and Systems theory	C	2
EEG 211	Fundamentals of Electrical Engineering I	C	2
EEG 219	Fundamentals of Electrical Engineering I Laboratory	C	1
CEG 211	Mechanics of Materials I	C	3
MEG 212	Fundamentals of Thermodynamics	C	2
MEG 211	Engineering Drawing	C	2
GST 201	General African Studies I	C	2
Total Units of Compulsory Courses			20
Total Units of Elective Courses			0

200L SECOND SEMESTER			
Course Code	**Course Title**	**Status C or E**	**Units**
GEG 222	Introduction to Engineering Statistics	C	3
GEG 228	Engineers in the Society	C	1
SSG 222	Introduction to Systems Methodology	C	1
SSG 226	Introductory Numerical Methods	C	2
SSG 227	Engineering Computer Graphics	C	2
CEG 221	Mechanics of Materials II	C	2
EEG 220	Fundamentals of Electrical Engineering II Laboratory	C	1
EEG 222	Fundamentals of Electrical Engineering II	C	2
EEG 224	Introduction to Switching and Logic Systems	C	2
MEG 222	Fundamentals of Fluid Mechanics	C	2
Total Units of Compulsory Courses			18
Total Units of Elective Courses			0

300L FIRST SEMESTER			
Course Code	**Course Title**	**Status C or E**	**Units**
GEG 311	Calculus of Several Variables	C	3
SSG 311	Mathematical Modeling for Artificial Intelligence Systems	C	3
SSG 317	Operations Research I	C	2
SSG 335	Introductory Robotic Systems	C	2
SSG 336	Robotics Laboratory I	C	1
EEG 315	Electronics Circuits I Laboratory	C	1
EEG 317	Instrumentation and Measurement	C	2
EEG 335	Electronics Circuits I	C	2
MEG 311	Mechanical Engineering Technology	C	3
GST 307	Entrepreneurship & Corporate Governance I	C	2
Total Units of Compulsory Courses			21
Total Units of Elective Courses			0

300L SECOND SEMESTER			
Course Code	**Course Title**	**Status C or E**	**Units**
GEG 322	Operational Methods I	C	3
SSG 320	Rigid Body Dynamics	C	2
SSG 321	Introduction to Continuum Mechanics	C	2
SSG 322	Control Theory I	C	2
SSG 324	Algorithms and Data Structures	C	2
SSG 326	Engineering Materials& the Environment	C	2
SSG 328	Operations Research II	C	2
SSG 340	Technical Computing Laboratory	C	1
SSG 342	Control Laboratory I	C	1
SSG 348	Industrial Engineering	C	2
Total Units of Compulsory Courses			19
Total Units of Elective Courses			0

400L FIRST SEMESTER			
Course Code	**Course Title**	**Status C or E**	**Units**
GEG 411	Technical Communications	C	1
GEG 412	Numerical Methods in Engineering	C	3
GEG 413	Engineering Statistics	C	3
SSG 411	Stochastic Models	C	2
SSG 415	Mechatronics and Robotic Systems	E	2
SSG 418	Systems Simulation I	C	2
SSG 431	Continuum Mechanics II	E	2
SSG 432	Statistical Quality Control	E	2
SSG 433	Operational Methods II	E	2
SSG 434	Embedded Systems Programming	C	2
SSG 435	Control Systems Analysis	E	2
SSG 437	Emerging Concepts and Technologies	C	1
SSG 438	Bio-Inspired Computing	E	2
SSG 439	Production Planning and Inventory Control	E	2
Total Units of Compulsory Courses			14
Total Required Units of Elective Courses			6

*LAB courses indicated as 'Electives' are compulsory for the respective specialization

Specializations

Engineering Analysis, Mechanics and Control: SSG 431, SSG 432, SSG 435

Industrial and Management Engineering: SSG 415, SSG 432, SSG 439

Artificial Intelligence and Mechatronics: SSG 415, SSG 435 SSG 438

400 LEVEL SECOND SEMSTER: SIW 400 (Industrial Training) – 6 Units

500L FIRST SEMESTER			
Course Code	**Course Title**	**Status C or E**	**Units**
ECN 451	Engineering Economics	C	2
SSG 511	Computer Graphics	C	2
SSG 513	Techniques of Planning and Scheduling	E	2
SSG 515	Mechatronics and Robotic Systems II	E	3
SSG 517	Systems Simulation II	C	3
SSG 518	Artificial Intelligence	E	2
SSG 530	Final Year Project I	C	3
SSG 531	Control Theory II	E	2
SSG 532	Engineering Design and Prototyping	C	2
SSG 533	Computational Methods I	E	3
SSG 535	Introduction to Machine Learning	E	2
SSG 537	Human Factors Engineering and Ergonomics	E	3
Total Units of Compulsory Courses			12
Total Units of Required Elective Courses			5

Specializations

Engineering Analysis, Mechanics and Control: SSG 531, SSG 533

Industrial and Management Engineering: SSG 513, SSG 537

Artificial Intelligence and Mechatronics: SSG 515, SSG 518 or SSG 535

500L SECOND SEMESTER			
Course Code	**Course Title**	**Status C or E**	**Units**
CIL 524	Introduction to Engineering Contract	C	1
BUS 430	Management for Engineers	C	1
SSG 521	Engineering Project Handling	E	1
SSG 522	Systems Analysis and Dynamics	C	2
SSG 523	Computational Methods II	E	3
SSG 524	Automated Reasoning	E	2
SSG 525	Facility Planning	E	2
SSG 526	Mechanics of Continua	E	2
SSG 527	Control of Robots and Human Arms	E	2
SSG 528	Manufacturing Systems Automation	E	2
SSG 529	Systems Animation	E	3
SSG 542	Image Processing	E	3
SSG 549	Systems Reliability and Maintainability	C	2
SSG 560	Final Year Project II	C	3
Total Units of Compulsory Courses			9
Total Units of Required Elective Courses			7

Specializations

Engineering Analysis, Mechanics and Control: SSG 523, SSG 526, SSG 528

Industrial and Management Engineering: SSG 525, SSG 528, SSG 529

Artificial Intelligence and Mechatronics: SSG 524, SSG 527 or SSG 528, SSG 542

SECTION 8a: Summary of number of units compulsory and elective courses to be taken/available at each Level for B.Sc. (Hons.)…Systems Engineering……

	First Semester		Second Semester			
Level	Units of Compulsory Courses	Units of Elective courses	Units of Compulsory Courses	Units l of Elective courses	Total of Compulsory Courses	Total of Elective courses
100	18	0	17	0	35	0
200	20	0	18	0	38	0
300	21	0	19	0	40	0
400	14	6	6	0	20	6
500	12	5	9	7	21	12
Total					154 (89.5%)	18 (10.5%)

SECTION 9: COURSE CONTENT

List the course content course by course indicating the course code, title, unit and the pre-requisite course code (if any) with the full content of the course. An example is given below

GEG 113 – ENGINEERING APPLIED MATHEMATICS I - 3 Units

Representation of vectors: Resultant of several vectors. Vectors in Euclidean space: lines, planes and spheres. The dot and cross products. Direction cosines. Differentiation of vector functions. Lami's theorem. Polygon of forces. Conditions for equilibrium of coplanar forces. Newton's laws of motion.Analytical treatment of static equilibrium of particles and rigid bodies. Distributed forces. Centroids and centres of gravity. Moments of Inertia. Analysis of structures and trusses. Forces in beams and tables. Friction.

GEG 115 – ENGINEERING ALGEBRA I - 2 Units

Axiomatic Set theory. Operations on Set. Boolean Algebra. Switching circuits, logic circuits and propositional logic. Transfinite induction and recursion. Sequences. Monotonic sequences and Convergence. Cauchy criteria. Series. Power series. Tests for convergence. Operations on power series.

GEG 117 – ENGINEERING CALCULUS I - 2 Units

Functions, limits, Continuity and Differentiability. Mean Value theorems. Techniques of differentiation for derivatives of algebraic, trigonometric, exponential and logarithmic functions; Curve sketching. L'Hospital rule. McLaurin's and Taylor's series. Applications of differentiation to rates of changes, maximum and minimum problems, application to rectilinear motion.

GEG 124 – ENGINEERING APPLIED MATHEMATICS II - 3 Units

An introduction to kinematics and kinetics of a particle. Systems of particles and rigid bodies. Energy and momentum methods. Applications. Impulsive motions. Motion of a rigid body (i) about a fixed axis (ii) in a plane. Equations of motion.